The Nose Knows:
Teaching Artificial Intelligence (AI) To Sniff Out Coffee And Beyond

Purpose

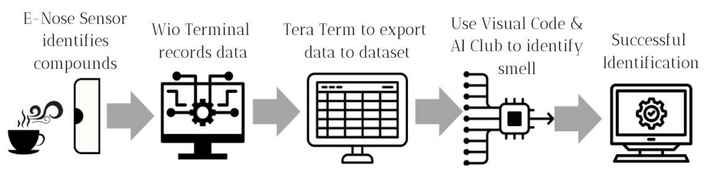
The purpose of this experiment is to test if the scientist can build an inexpensive nose prototype that will collect data from coffee samples and air that can be used to train artificial intelligence (AI) on that data to identify the smell. Future research can then be used to transition to detecting volatile organic compounds (VOCs) present in the Cancer Odor Database (COD) for a future experiments.

Background

The investigation was inspired by a combination of articles on the ability of dogs to sniff cancer and the artificial nose experiment by Benjamin Cabe. A way to remove the tumor that causes lung cancer is possible. The hard part is early detection before the tumor kills the lung.

The scientist detected smell using Grove. The scientist wanted to expand this experiment utilizing the COD database as recognition for the smell sensors but initially recognize coffee first and then transition to the database.

The scientist first collected the data of air as the control variable and measured the nitrogen dioxide ($N0_2$), carbon monoxide (CO), Ethanol, (C_2H_5OH) and total volatile organic compounds (VOCs). The scientist than collected the same data for coffee by placing the cup next to the sensor. This process was repeated 15 times to create a reliable database. It was important to move to a different room when doing air after coffee as the smell is strong and stayed in the air.



Impact

According to the American Cancer Society, lung cancer is the second most common type of cancer and is responsible for more than 130,180 deaths(2). Early detection improves treatment choices with less aggressive options which improve the quality of life and significantly reduce mortality rates. The impact of this experiment shows that it is possible to detect certain compounds using a variety of gas sensors to collect data that can be used to create artificial intelligence (AI) models that would correctly detect the scent.

Research Question

Can a scientist develop a methodology that can be used to detect a mixture of inorganic compounds (nitrogen, carbon dioxide, etc) and VOCs to identify scent?

Hypothesis

The hypothesis is that the learning algorithm can be trained to measure the $N0_2$, CO, C_2H_5OH and total VOCs of coffee to correctly identify scent.

Materials and Tools

- MOSFET board
- Gas Sensor
- Connector
- Terminal Batteries
- Wio Terminal
- Data from COD
- Computer
- Charger
- Cables
- Screws For Assembly
- Fan
- Jumper wires
- Chair
- Python Coding
- AI Club
- Tera term
- Wifi/Internet
- Tinkercad
- AutoCAD
- USB cable
- PlatformIO
- 3D Printer

Methods

Safety Protocol:

1. Ensure all cords and plugs free of broken insulation, exposed wiring
2. Check that no electrical outlets are overloaded
3. Ensure to use a GFCI protected outlet
4. Make sure the battery charger is situated so there is air circulating around it
5. Double check that terminals are not bent, and no cracks in casing
6. Wear Blue Light glasses when working on the computer.
7. Pay close attention when handling any hot liquids
8. Ensure that there is a adult present

Procedures:

Step 1: Make the Nose: The Cane Institute for Advanced Technologies will print the nose enclosure on the researchers behalf
- 3D print the nose enclosure from thingiverse.com/thing:4493907

Step 2: Connect all the Electronics: Connect Wio Terminal to the gas sensors which are connected to the I^2C interface, and the MOSFET is connected to digital output D0. Use short M2 screws with washers and nuts to attach the Wio Terminal and the DC fan to the back of the nose enclosure

Step 3: program the Nose: Use PlatformIO to compile and upload the update firmware.The Wio Terminal should start in Training mode to show a real-time graph of the various gases being detected from the coffee smells.

Step 4: Turn it on 24 hours before using to heat internal sensors, calibrate & Train AI nose: We are getting the data from our device and making a dataset out of it and using that dataset to train my AI.

Step 5: We are giving the sensor data to the AI to see if what it is detecting is air or coffee

Step 6: Test using coffee and adjust code as needed

Disposal Protocol:

1. Place all hardware into designated box
2. Turn off all electronics
3. Take stretch breaks to ensure no computer related injuries from overuse
4. Pour out all liquids in the sink



3D printed nose which holds all sensors sitting on top of coffee.

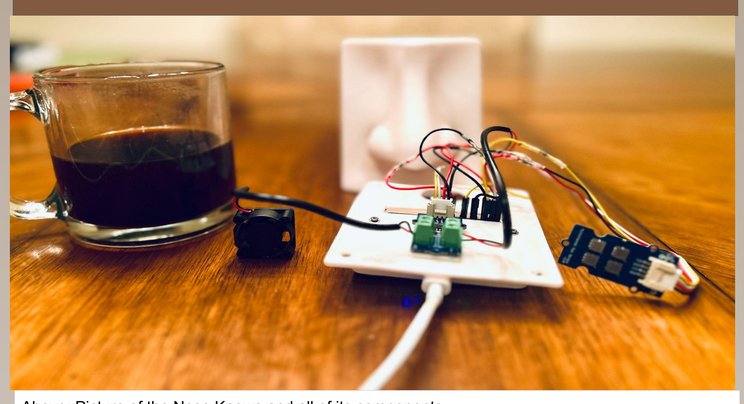
Data Error Matrix

		Reality: Coffee	Reality: Not Coffee
Tested	Coffee	CORRECT (100)	False Positive Type 1 error (0)
Tested	Not Coffee	False Negative Type 2 error (0)	CORRECT (100)

Variables/Controls

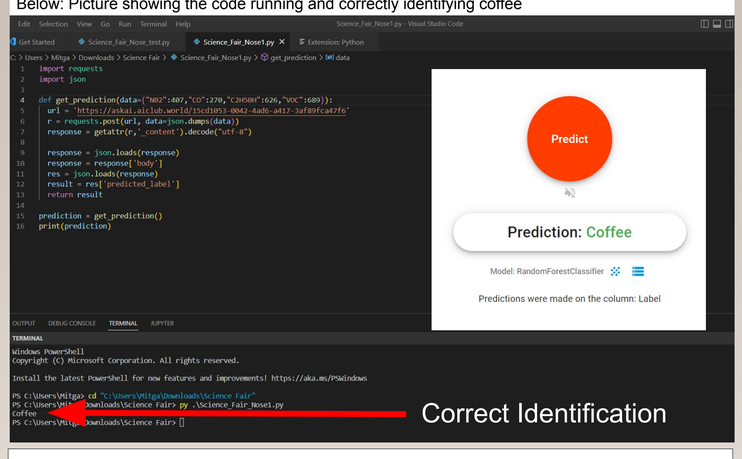
The scientist used air as the control variable and coffee as the variable scent.

Data Collection



Above: Picture of the Nose Knows and all of its components

Below: Picture showing the code running and correctly identifying coffee



The scientist collected and recorded N02, CO, C2H5OH and VOCs for coffee for two minutes and than air for two minutes 100 times resulting in over 120,000 data points for each. In order to do this, the scientist had to train the AI using data sets which was done by running multiple different scents in front of the sensor and then using a fan to clear it before putting another scent. The artificial nose was built using a template by Mr. Cabe and then was used to collect data on air and coffee. It is coded to automate the process of identifying if the scent is coffee or air. This same process can be used in the future to measure lung cancer breath.

Results

Chart of Air Compounds

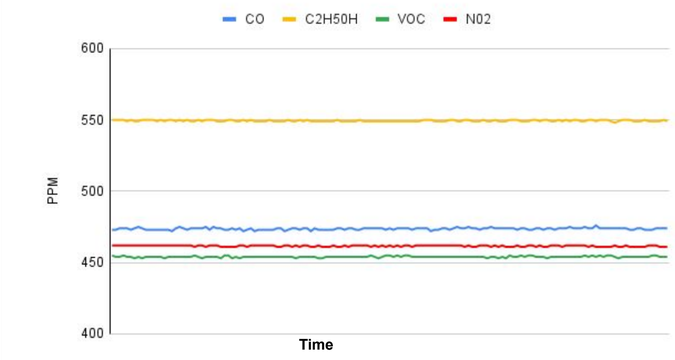
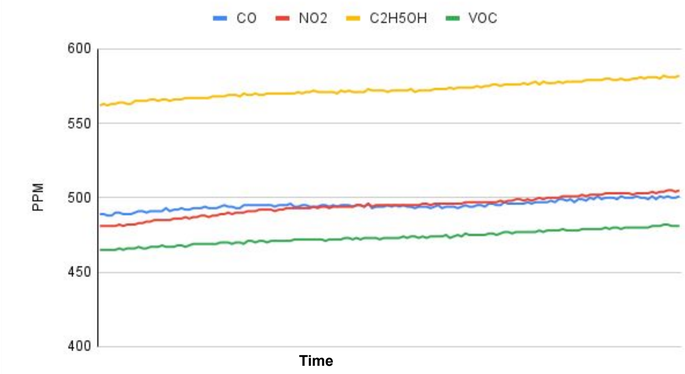


Chart of Coffee Compounds



The nose was able to correctly identify the presence of certain compounds in scent using coffee. The scientist was able to collect data and use that data to create an AI to properly identify the scent based on a mixture of compounds (nitrogen, carbon dioxide, etc) and VOCs to correctly identify the scent of coffee. The scientist repeated the experiment many times and was able to replicate the correct identification each time.

Data Analysis

The difference in composition on average for coffee vs. air is 33 parts per million (parts per million) nitrogen dioxide ($N0_2$), 21 parts per million carbon monoxide (CO), 23 parts per million Ethanol (C_2H_5OH) and 19 parts per million total volatile organic compounds (VOCs). Despite this small difference, the software program was able to successfully identify coffee correctly the 30 times the scientist ran the experiment. This shows that the technology works smelling complex compounds some of which are the same in human breath and could be used to identify lung cancer. The data shows there is a difference.

Validity and Reliability

The test was run over 100 times to successfully identify coffee vs air and was successful each time suggesting it 100% reliable.

Conclusion

The scientist set out to design a program that would smell coffee as a temporary replacement for lung cancer automaton model within to detect volatile organic compounds. Using artificial intelligence, the scientist wanted to be able to identify different compounds in coffee as a temporary replacement to show that their technology could work.

The scientist was successful in creating a model that can be used to positively identify the presence of certain compounds in scent using coffee. The scientist was able to collect data and use that data to create an AI to properly identify the scent based a mixture of inorganic compounds (nitrogen, carbon dioxide, etc) and VOCs to correctly identify the scent of coffee.

Future Applications

- Replicate this model using samples of lung cancer breath to assist with early detection. A database would have to be created from lung cancer breath samples similar to how the scientist did with coffee.

- It could also be done for other cancer types like the chronic Obstructive Pulmonary Disease which already have datasets that could be used to train the AI.

References

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All photos, tables, graphs, and drawings created by the researcher, Anik Sahai, unless otherwise noted.